Bin packing and cutting stock

Outline

Bin packing

Introduction: bin packing

- Items of given weights must be packed into identical bins of capacity c.
- The goal is to use as few bins as possible.
- The problem appears in loading, storage and allocation to identical servers.

Definition 1: bin packing problem

Given items $\mathscr{I}$ with weights $w_j \leq c$, partition them into the minimum number of subsets $S_1, \ldots, S_q$ such that

$$\sum_{j \in S_i} w_j \leq c, \qquad i = 1, \ldots, q.$$

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Bounds

- Any feasible packing with q bins is a primal bound.
- The total weight gives the dual bound

$$\left\lceil \frac{\sum_{j \in \mathscr{J}} w_j}{c} \right\rceil \leq z(I).$$

- Additional combinatorial arguments can strengthen this lower bound.

Summary

Bin packing partitions items into interchangeable bins. It is a minimization problem: a packing gives an upper bound and a lower bound certifies that fewer bins cannot suffice.

Cutting stock

Introduction: cutting stock

Cutting stock cuts identical rolls, bars or sheets into required pieces with minimum waste or minimum stock material.

Definition 2: cutting stock problem

Given stock pieces of length c and demands for pieces of lengths w_j, choose cutting patterns and their multiplicities so that every demand is met with a minimum number of stock pieces.

Relation with bin packing

- Bin packing assigns items to bins; cutting stock assigns demanded pieces to cutting patterns.
- A pattern is a feasible multiset of pieces in one stock roll.
- The two problems share capacity constraints but differ in their objective and demand requirements.

Typical mistakes

Typical mistakes

- Counting differently labelled bins as different packings.
- Confusing a feasible packing with a proof that it uses the minimum number of bins.
- Forgetting that cutting stock must meet every demand.

Summary

- Bin packing minimizes bins; cutting stock satisfies demand with cutting patterns.
- Both are packing problems and both need primal and dual bounds.